A Practical Guide to Naive Bayes classifier

TOPIC -- NAIVE BAYES CLASSIFIER

$$P(C_k \mid x) \propto P(C_k) \prod_{i=1}^{n} P(x_i \mid C_k)$$

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y ^ = argmax k ∈ { 1 , … , K } p (C k) ∏ i = 1 n p (x i ■ C k) . {\displaystyle {\hat {y}}={\underset {k\in \{1,\ldots ,K\}}{\operatorname {argmax} }}\ p(C_{k})\displaystyle \prod _{i=1}^{n}p(x_{i}\mid C_{k}).}

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The following example assumes equiprobable classes so that P(male)= P(female) = 0.5. This prior probability distribution might be based on prior knowledge of frequencies in the larger population or in the training set.

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where the evidence Z = p (x) = ∑ k p (C k) p (x ■ C k) {\displaystyle Z=p(\mathbf {x})=\sum _{k}p(C_{k})\ p(\mathbf {x} \mid C_{k})} is a scaling factor dependent only on x 1 , … , x n {\displaystyle x_{1},\ldots ,x_{n}} , that is, a constant if the values of the feature variables are known. Often, it is only necessary to discriminate between classes. In that case, the scaling factor is irrelevant, and it is sufficient to calculate the log-probability up to a factor: ln ■ p (C k ■ x 1 , … , x n) = ln ■ p (C k) + ∑ i = 1 n ln ■ p (x i ■ C k) − ln ■ Z ■ irrelevant {\displaystyle \ln p(C_{k}\mid x_{1},\ldots ,x_{n})=\ln p(C_{k})+\sum _{i=1}^{n}\ln p(x_{i}\mid C_{k})\underbrace {-\ln Z} _{\text{irrelevant}}} The scaling factor is irrelevant, since discrimination subtracts it away: ln ■ p (C k ■ x 1 , … , x n) p (C l ■ x 1 , … , x n) = (ln ■ p (C k) + ∑ i = 1 n ln ■ p (x i ■ C k)) − (ln ■ p (C l) + ∑ i = 1 n ln ■ p (x i ■ C l)) {\displaystyle \ln {\frac {p(C_{k}\mid x_{1},\ldots ,x_{n})}{p(C_{l}\mid x_{1},\ldots ,x_{n})}}=\left(\ln p(C_{k})+\sum _{i=1}^{n}\ln p(x_{i}\mid C_{k})\right)-\left(\ln p(C_{l})+\sum _{i=1}^{n}\ln p(x_{i}\mid C_{l})\right)} There are two benefits of using log-probability. One is that it allows an interpretation in information theory, where log-probabilities are units of information in nats. Another is that it avoids arithmetic underflow.

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To estimate the parameters for a feature's distribution, one must assume a distribution or generate nonparametric models for the features from the training set. The assumptions on distributions of features are called the "event model" of the naive Bayes classifier. For discrete features like the ones encountered in document classification (include spam filtering), multinomial and Bernoulli distributions are popular. These assumptions lead to two distinct models, which are often confused.

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p (x i ■ x i + 1 , … , x n , C k) = p (x i ■ C k) . {\displaystyle p(x_{i}\mid x_{i+1},\ldots ,x_{n},C_{k})=p(x_{i}\mid C_{k})\,.}

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Thus, the probability ratio p(S | D) / p(¬S | D) can be expressed in terms of a series of likelihood ratios. The actual probability p(S | D) can be easily computed from log (p(S | D) / p(¬S | D)) based on the observation that p(S | D) + p(¬S | D) = 1. Taking the logarithm of all these ratios, one obtains:

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The link between the two can be seen by observing that the decision function for naive Bayes (in the binary case) can be rewritten as "predict class C 1 {\displaystyle C_{1}} if the odds of p (C 1 ■ x) {\displaystyle p(C_{1}\mid \mathbf {x})} exceed those of p (C 2 ■ x) {\displaystyle p(C_{2}\mid \mathbf {x})} ". Expressing this in log-space gives:

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Disadvantages Depending on the implementation, Bayesian spam filtering may be susceptible to Bayesian poisoning, a technique used by spammers in an attempt to degrade the effectiveness of spam filters that rely on Bayesian filtering. A spammer practicing Bayesian poisoning will send out emails with large amounts of legitimate text (gathered from legitimate news or literary sources). Spammer tactics include insertion of random innocuous words that are not normally associated with spam, thereby decreasing the email's spam score, making it more likely to slip past a Bayesian spam filter. However, with (for example) Paul Graham's scheme only the most significant probabilities are used, so that padding the text out with non-spam-related words does not affect the detection probability significantly. Words that normally appear in large quantities in spam may also be transformed by spammers. For example, «Viagra» would be replaced with «Viaagra» or «V!agra» in the spam message. The recipient of the message can still read the changed words, but each of these words is met more rarely by the Bayesian filter, which hinders its learning process. As a general rule, this spamming technique does not work very well, because the derived words end up recognized by the filter just like the normal ones. Another technique used to try to defeat Bayesian spam filters is to replace text with pictures, either directly included or linked. The whole text of the message, or some part of it, is replaced with a picture where the same text is "drawn". The spam filter is usually unable to analyze this picture, which would contain the sensitive words like «Viagra». However, since many mail clients disable the display of linked pictures for security reasons, the spammer sending links to distant pictures might reach fewer targets. Also, a picture's size in bytes is bigger than the equivalent text's size, so the spammer needs more bandwidth to send messages directly including pictures. Some filters are more inclined to decide that a message is spam if it has mostly graphical contents. A solution used by Google in its Gmail email system is to perform an OCR (Optical Character Recognition) on every mid to large size image, analyzing the text inside.

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Semi-supervised parameter estimation Given a way to train a naive Bayes classifier from labeled data, it's possible to construct a semi-supervised training algorithm that can learn from a combination of labeled and unlabeled data by running the supervised learning algorithm in a loop: